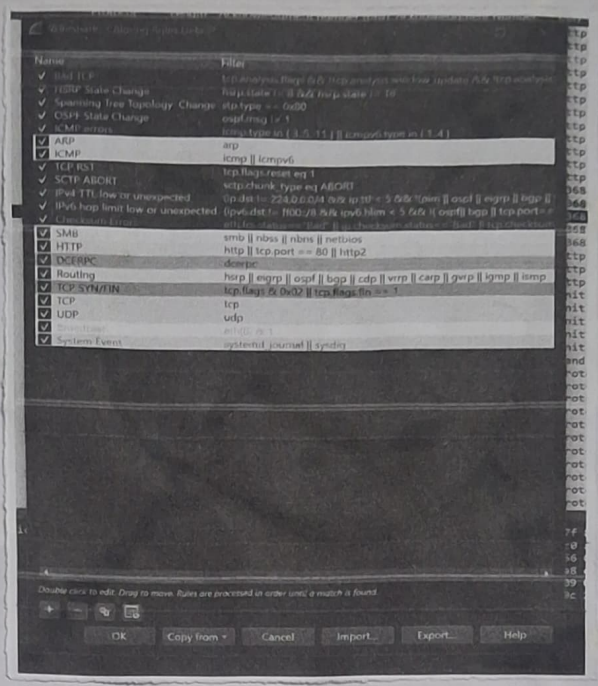
Practical - 5

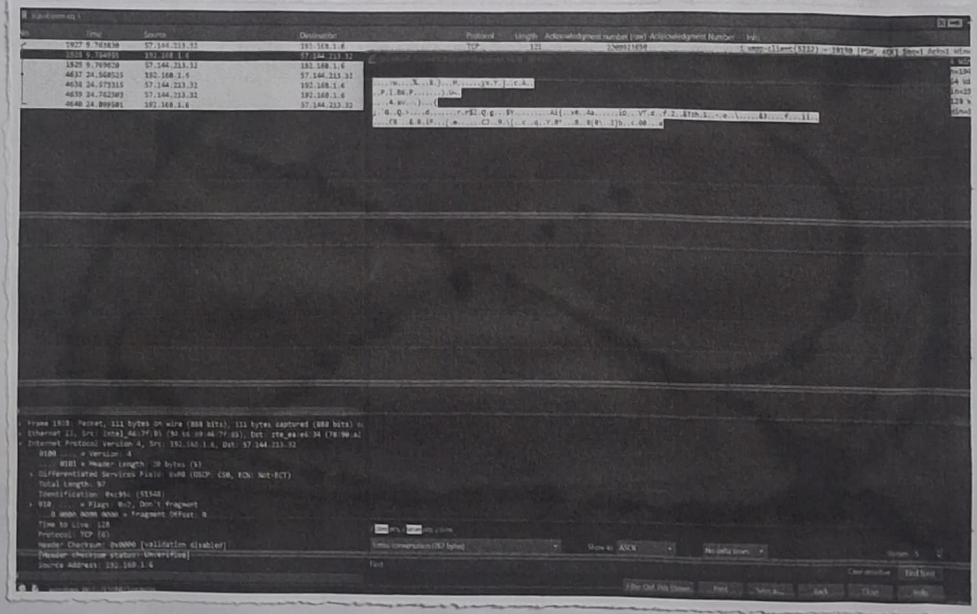
Aim :-
Experiments on packet capture tool wireshark

Packet Sniffer :-
→ Sniffs messages being sent/received from/by your computers.
→ Store and Display the contents of the various protocol fields in the message
→ passive
i) Never sends packets itself
ii) No packets adds to it
iii) Receivers a copy of all packets (sent/Received)

Packet Sniffer structure Diagnostic tool :-
*) Tcpdump - Eg:- tcpdump -epx host 10.129.41.2 -w exe3.out
*) Wire shark - Wire Shark -r exe3.out .

Procedure :-
→ Select the local area connection in wireshark.
→ Go to Capture options
→ Select capture and Stop at automatically after 100packets
→ Then click start Capture
→ Save packets


Name
Filter
ARP
arp
ICMP
icmp || icmpv6
TCP RST
tcp.flags.reset eq 1
SCTP ABORT
sctp.chunk_type eq ABORT
SMB
smb || nbss || nbns || netbios
HTTP
http || tcp.port == 80 || http2
DCERPC
dcerpc
Routing
hsrp || eigrp || ospf || bgp || cdp || vrrp || carp || gvrp || igmp || ismp
TCP SYN/FIN
tcp.flags & 0x02 || tcp.flags.fin == 1
TCP
tcp
UDP
udp
System Event
systemd_journal || syslog
Double click to edit. Drag to move. Rules are processed in order until a match is found.
OK
Copy from
Cancel
Import...
Export...
Help



i) Create a filter to display only TCP/IP packets, inspect the packets and provide the flow graph

→ Go to search bar at the top of the tool.
→ Search TCP packets in search bar.
→ To see flow graph click statistics flow graph.
→ Save the packets

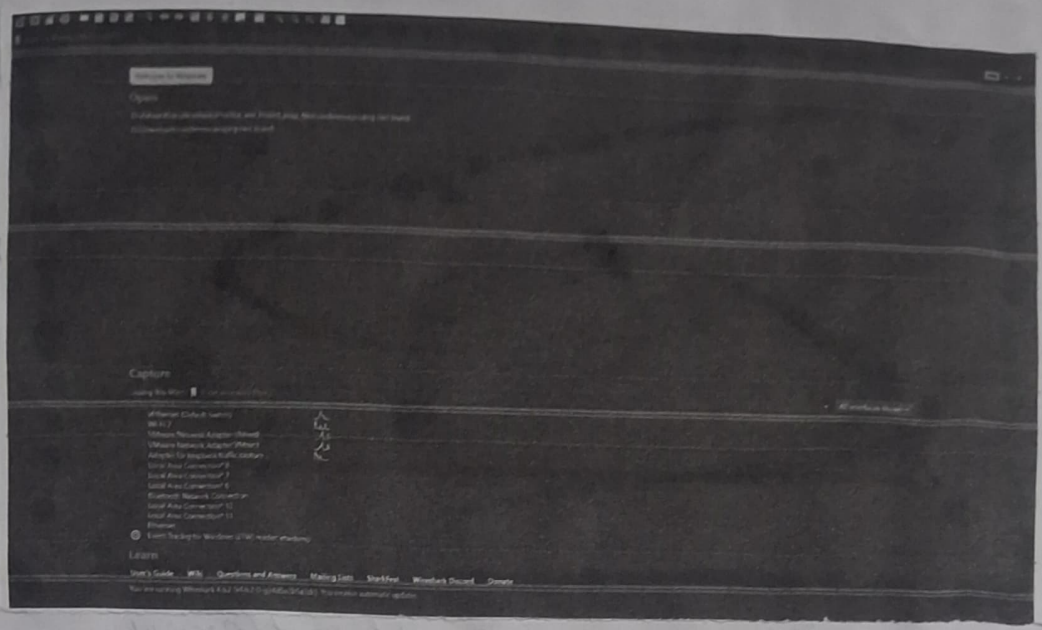
ii) Create a filter for ARP packets

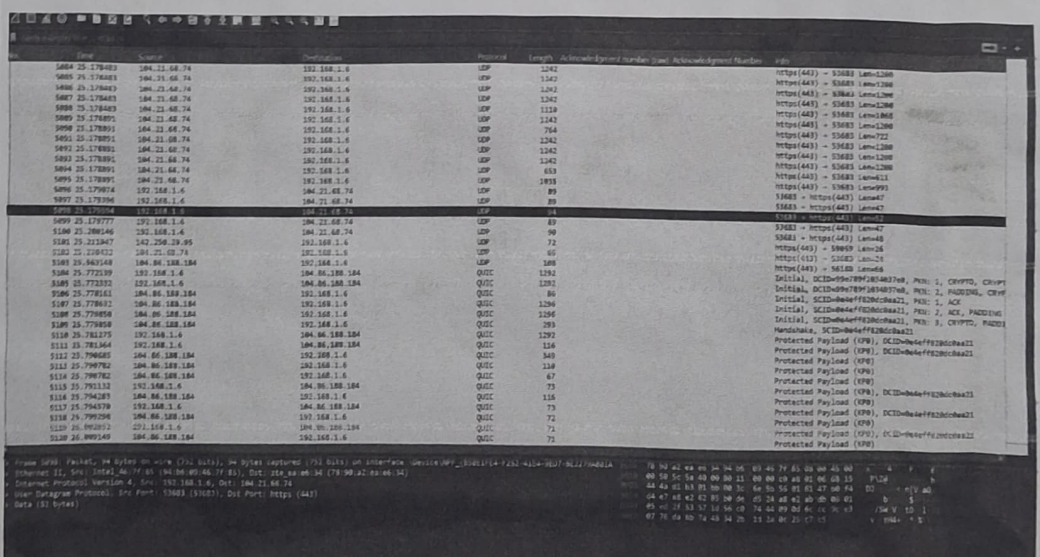
* Go to search option
* Search ARP packets in search bar
* Save the packets

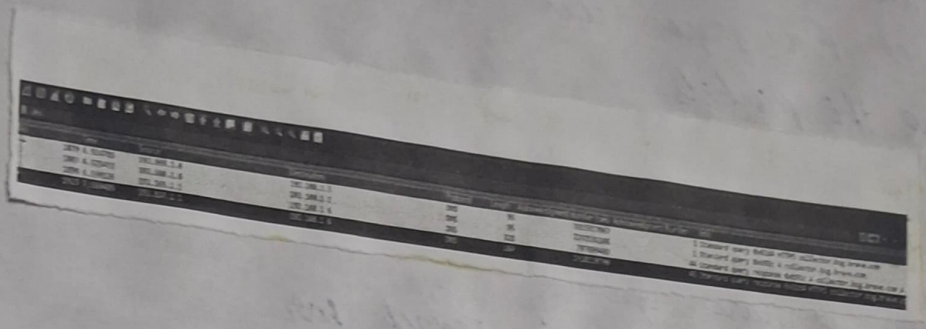
iii) Creating a filter to display DNS packets and provide the flow graph.

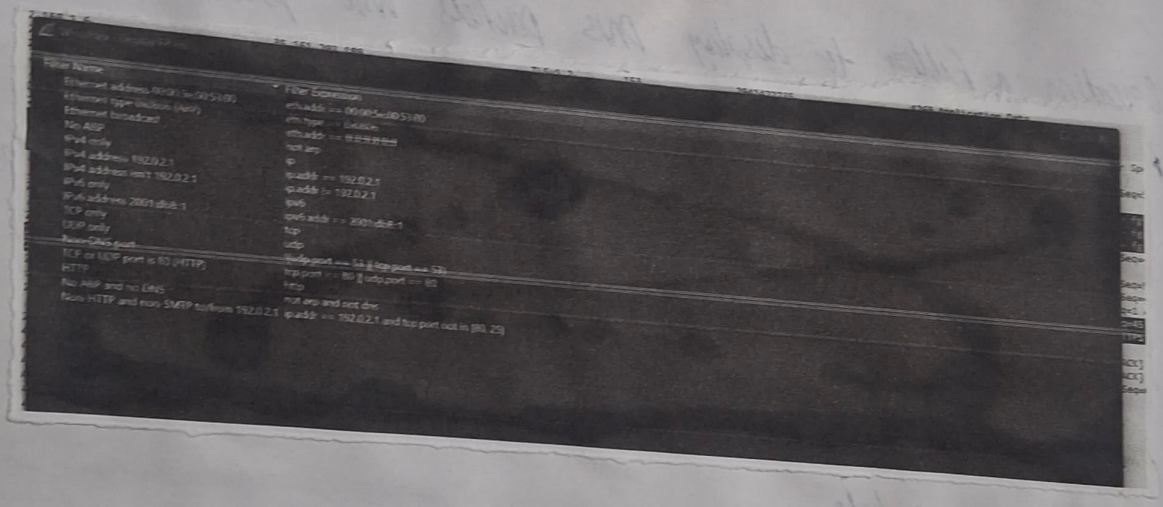
iv) Creating a filter for ICMP and DHCP packets and inspect and show the packets

v) Create filter for ICMP and DHCP packets and inspect and show the packets


Capture
Learn






Filter Name
Filter Expression
Ethernet address 00:08:5e:00:53:00
eth.addr == 00:08:5e:00:53:00
Ethernet type 0x0806 (ARP)
arp
Ethernet broadcast
eth.addr == ff:ff:ff:ff:ff:ff
No ARP
not arp
IPv4 only
ip
IPv4 address 192.0.2.1
ip.addr == 192.0.2.1
IPv4 address != 192.0.2.1
ip.addr != 192.0.2.1
IPv6 only
ipv6
IPv6 address 2001:db8::1
ipv6.addr == 2001:db8::1
TCP only
tcp
UDP only
udp
TCP or UDP port is 80 (HTTP)
HTTP
tcp.port == 80 || udp.port == 80
No ARP and no DNS
http
not arp and not dns
Non-HTTP and non-SMTP to/from 192.0.2.1
ip.addr == 192.0.2.1 and !(tcp.port in {80 25})

Result:-

Wireshark help us in analysing and understanding ~~the~~ our network traffic and packet flow through various filters and flow graphs.